SHOPIFY SALES & CUSTOMER ANALYTICS

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ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Applied Predictive Analytics (22ADE57N.PE2)

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CERTIFICATE

This is to certify that the course end project report **SHOPIFY SALES & CUSTOMER ANALYTICS** (Applied Predictive Analytics) being submitted by

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in partial fulfillment for the award of the Degree of Bachelor of Engineering in Artificial Intelligence and Data Science to the CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD carried out under my mentorship and supervision during the year 2026-27. The results presented in this report have not been submitted to any other university or Institute for the award of any degree.

Mrs. V. Vasuki Rohini Devi
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ABSTRACT

This project analyses Shopify e-commerce transaction data to understand sales performance, product contribution, customer purchasing behaviour and repeat-purchase activity. The dataset contains 7,431 transaction records from 18 March 2025 to 24 March 2025, representing 7,431 distinct orders and 4,431 customers. Power BI is used to develop an interactive, three-page business intelligence report, while Python is used for data preparation, exploratory analysis, visualisation and a supervised machine-learning experiment.

The analysis reports total revenue of USD 4,598,960.87 and an average order value of USD 618.89. Running Shoes is the leading product type by revenue, contributing approximately USD 1.62 million. The observed-period repeat-customer rate is 46.02%, and the average first-to-second purchase gap is 2.03 days according to the Power BI analysis. Logistic Regression and Random Forest are evaluated for predicting whether a customer places a second distinct order within three days of their first observed order. On the later-date test set, Random Forest obtains an F1-score of 0.37 and ROC-AUC of 0.49, indicating weak discrimination. The results therefore demonstrate the value of descriptive analytics while highlighting the limitations of building predictive models from a short observation window.

Keywords: *Shopify, e-commerce analytics, Power BI, customer behaviour, customer retention, classification, Logistic Regression, Random Forest.*

1. INTRODUCTION

1.1 Project Overview and Need

This project examines transaction-level data from a Shopify store to analyse revenue, orders, product performance and customer purchasing behaviour. Power BI presents the business metrics through interactive dashboards, while Python is used for data preparation, exploratory analysis and a supervised classification experiment.

The dataset covers 18–24 March 2025 and contains 7,431 records with order and customer identifiers, invoice dates, product types, quantities, transaction values, billing locations and payment gateways. Raw transaction records do not directly show which products drive revenue, how customers distribute across purchase-frequency groups, or how retention can be improved; this study consolidates these indicators for store owners, sales teams and customer-retention teams. As the dataset spans only seven calendar days, the findings describe this observation window and do not establish seasonality, long-term retention or production-grade predictive capability.

1.2 Objectives of the Project

- Measure total revenue, distinct orders, average order value and unique customers.
- Identify the product types that contribute most to revenue and order volume, and analyse daily revenue and order-count variation.
- Examine one-time and repeat customers, purchase frequency, customer revenue contribution and the observed first-to-second purchase gap.
- Develop interactive Power BI dashboards using a structured data model and DAX measures, and prepare the data and perform exploratory analysis using Python.
- Compare Logistic Regression and Random Forest for three-day repeat-purchase classification, interpret their performance, and state recommendations and limitations.

1.3 Scope of the Project

The project covers the supplied Shopify dataset, Power Query preparation, a Power BI star-schema-style model, DAX-based KPIs, Python exploratory analysis and an exploratory machine-learning task. The prediction target is whether a customer makes a second distinct order within three calendar days of the first observed order.

2. PROBLEM AND BUSINESS ANALYSIS

2.1 Business Problem Identification

Store teams that examine only individual transaction records have limited visibility of product contributions, daily sales, one-time versus repeat customers and higher-value customers. The project addresses this gap through an integrated dashboard and an exploratory predictive workflow. The expected impact is clearer sales and customer indicators, more focused review of high-contribution product types and a better basis for customer-engagement decisions. The machine-learning component is treated as an experiment whose usefulness is judged by test performance rather than assumed.

2.2 Stakeholders

Table 1. *Stakeholders, information needs and use of the analysis.*

Stakeholder	Information need	Use of the analysis
Store owner / management	Overall revenue, orders, AOV and product contribution	Monitor performance and plan business priorities
Sales team	Daily sales, order volume and product performance	Review sales movement and product demand
Marketing team	Customer mix, purchase frequency and customer value	Plan campaigns and relevant product communication
Customer-retention team	Repeat customers and purchase-gap indicators	Design follow-up and re-engagement activity
Inventory / operations team	Revenue and order contribution by product type	Review stock availability for leading categories

3. DATASET DESCRIPTION AND DATA PREPARATION

3.1 Dataset Overview and Attributes

The Excel workbook contains a sheet named “shopify_sales” with 7,431 rows and 19 columns, covering 18 March 2025 to 24 March 2025. The main fields used are Order Number, Customer Id, Invoice Date, Product Type, Quantity, Subtotal Price, Total Price Usd, Total Tax, CITY, Billing Address Province and Gateway; additional fields include billing country, currency, postal code, product identifier and variant identifier.

Table 2. *Attribute groups of the Shopify dataset.*

Attribute group	Representative fields	Analytical purpose
Transaction identifiers	Order Number, Customer Id	Count distinct orders and customers
Time	Invoice Date	Daily aggregation and purchase sequence
Product	Product Type, Product Id, Variant Id, Quantity	Product contribution and units
Financial	Subtotal Price, Total Price Usd, Total Tax	Revenue and order-value calculations
Location	CITY, Billing Address Province, Billing Address Country	Geographic grouping
Payment	Gateway, Currency	Payment-method context

3.2 Data Quality and Preprocessing

The quality check reported no exact duplicate rows. Product Id contained 11 missing values (approximately 0.15%) and Variant Id contained 4 (approximately 0.05%); neither field was required for the core customer-level classification. The preprocessing steps were:

- Removed exact duplicate rows if present (none in the supplied data) and records missing essential order, customer, date or revenue values.
- Converted Invoice Date to a datetime field and financial/quantity columns to numeric types.
- Standardised order and customer identifiers as trimmed strings.
- Replaced missing product, city, province and gateway labels with "Unknown" where those fields were used.

All 7,431 rows remained after preprocessing; the data ranged from 18 March 2025 00:00 to 24 March 2025 23:56.

3.3 Order-Level Data and Feature Engineering

Records were grouped by Customer Id and Order Number to create an order-level dataset with the earliest invoice timestamp, total order revenue, total quantity, a representative product type, number of distinct product types, city, province and payment gateway; date-derived fields included order day, weekday and order hour. Customers with more than one distinct order during the seven-day period were classified as repeat customers and the rest as one-time customers. This classification is specific to the observation window and is not a lifetime customer classification.

4. EXPLORATORY DATA ANALYSIS

4.1 Sales and Daily Performance

Table 3 summarises the key sales metrics. The daily variation shown in Figure 1 describes the observed week only and is not evidence of a recurring weekly or seasonal pattern.

Table 3. *Key sales metrics for 18–24 March 2025.*

Metric	Observed result
Transaction rows after cleaning	7,431
Distinct orders	7,431
Unique customers	4,431
Total revenue	USD 4,598,960.87
Average order value	USD 618.89
Lowest daily revenue	USD 578,282.08 (18 March 2025)
Highest daily revenue	USD 752,227.32 (24 March 2025)
Highest daily order count	1,204 (24 March 2025)

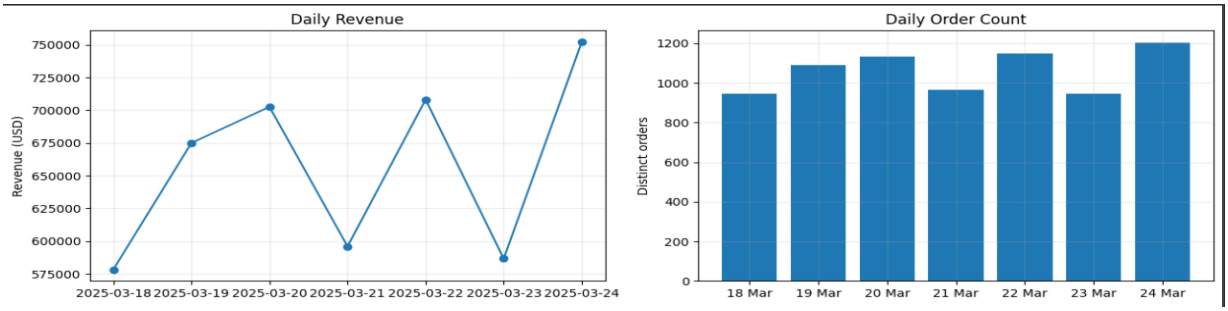


Figure 1. *Daily revenue and distinct order count, 18–24 March 2025.*

4.2 Product Performance

Running Shoes is the leading product type, contributing 35.16% of total product revenue; the top three categories together account for approximately 71.25% of recorded revenue, indicating that revenue is concentrated in a small number of leading product types (Table 4, Figure 2).

Table 4. *Top product types by revenue.*

Product type	Revenue (USD)	Revenue share
Running Shoes	1,616,946.28	35.16%
Tennis Shoes	971,943.82	21.13%
Walking Shoes	688,160.68	14.96%
Cycling Shoes	529,320.28	11.51%
Climbing Shoes	373,905.43	8.13%

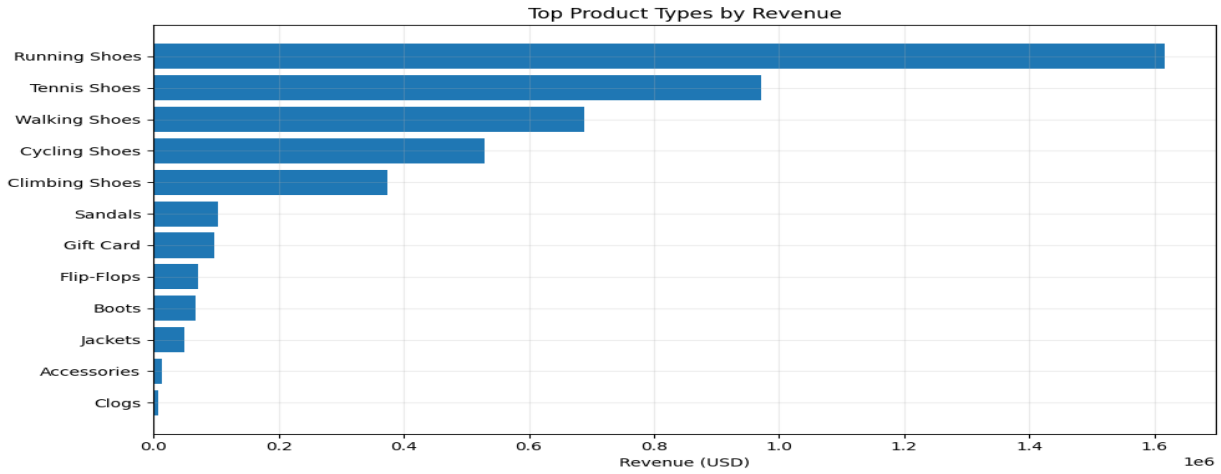


Figure 2. *Top product types ranked by revenue.*

4.3 Customer Behaviour and Retention

Of the 4,431 unique customers, approximately 2,390 were one-time customers (53.98%) and 2,041 were repeat customers (46.02%); the order-frequency distribution is concentrated at one order per customer (Figure 3). This indicates an opportunity to encourage a second purchase, although the short observation window may classify customers who purchase later as one-time customers. The Power BI analysis reports an average first-to-second purchase gap of 2.03 days among customers with an observed second purchase, and a top 10% revenue share of 22.05%. The cohort view contains only a March cohort because the records cover a single week.

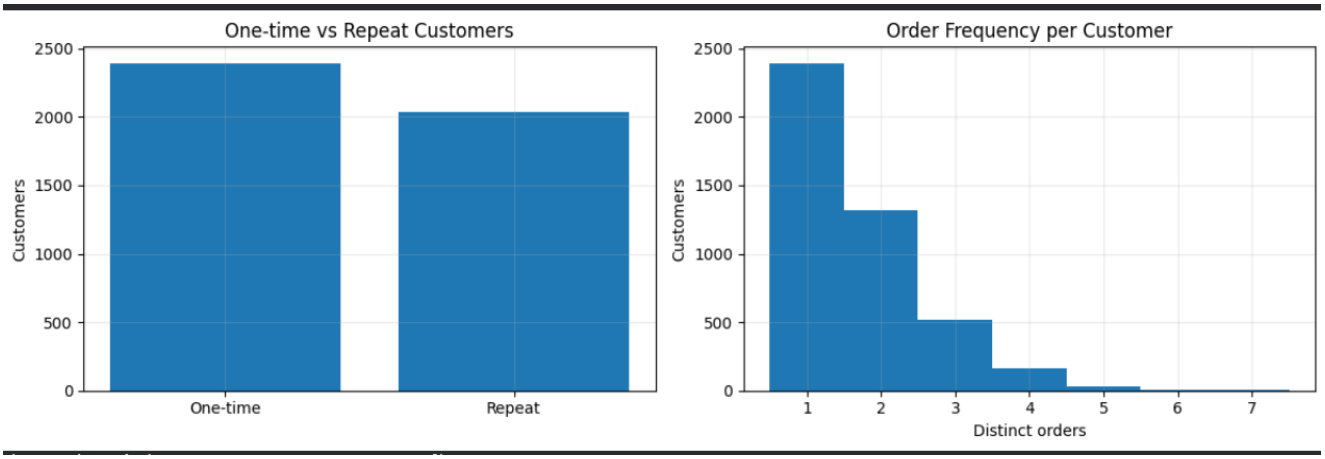


Figure 3. *One-time versus repeat customers and the distribution of order frequency.*

5. POWER BI DASHBOARD DEVELOPMENT

5.1 Data Model and Measures

The model follows a star-schema-style design with FactSales as the central transaction table and DimCustomer, DimProduct and DimDate as supporting dimensions; the dedicated DimDate table supports consistent date filtering and time-based calculations. Power Query was used to review and prepare data types, missing values, duplicates and standardised labels. DAX measures were created for Total Revenue, Total Orders, Average Order Value, Unique Customers, Repeat Customers, New Customers, Repeat Purchase Rate, Customer Revenue, Top 10% Customer Revenue, Top 10% Revenue Share, Average First-to-Second Purchase Gap and Revenue Growth.

5.2 Dashboard Pages

Table 5. *Power BI dashboard pages.*

Dashboard page	Main contents
Executive Dashboard	Headline KPIs for total revenue, orders, AOV and unique customers; high-level sales overview.
Sales & Customer Analysis	Product-type revenue, daily sales, customer mix, order frequency and customer contribution.
Customer Retention & Analysis	Repeat-purchase rate, purchase gap, cohort view and customer-value indicators.

Date, Product Type and Customer Type slicers were tested to support interactive exploration. Figures 4–6 show the three dashboard pages.

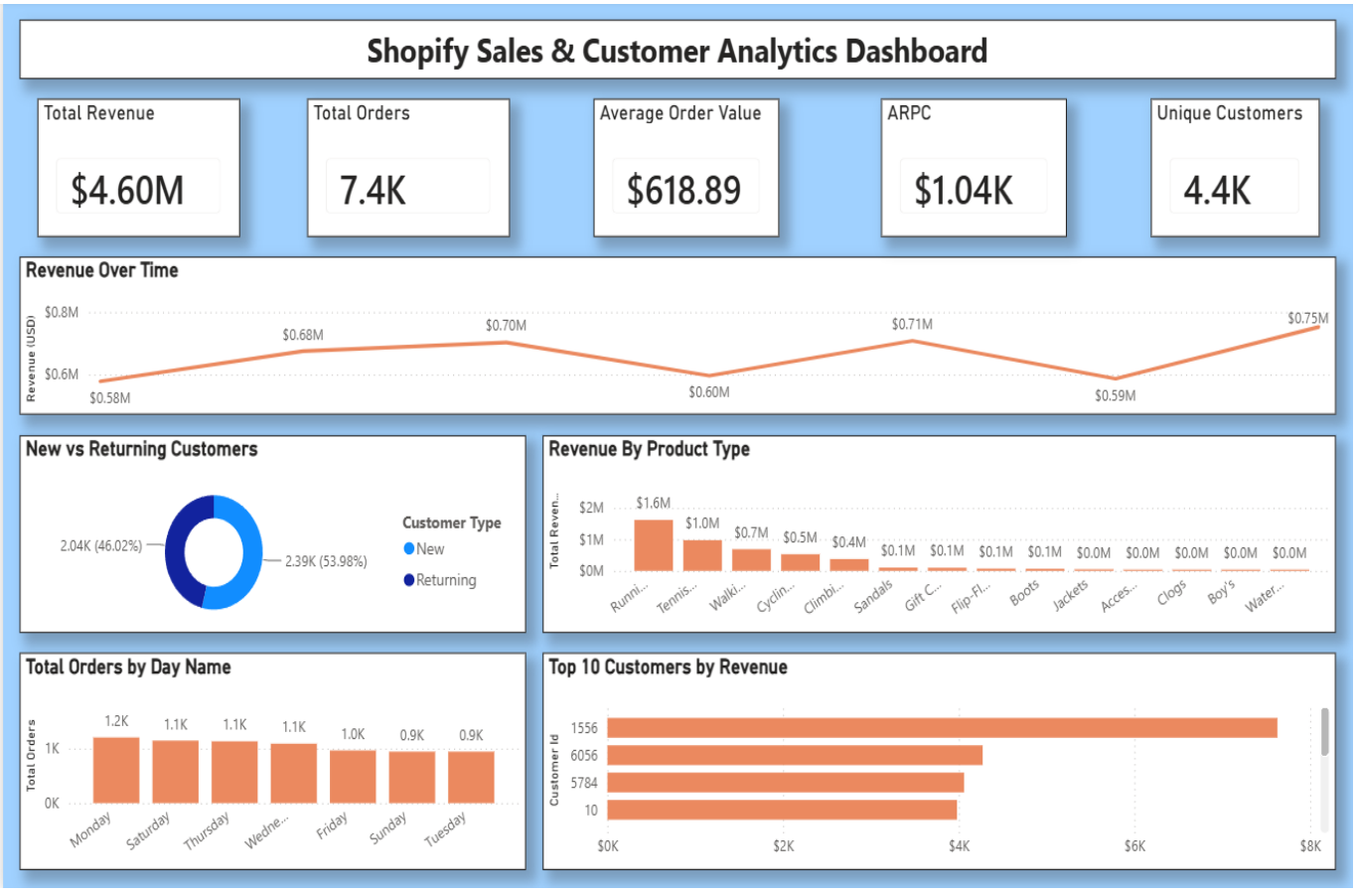


Figure 4. *Power BI Executive Dashboard.*

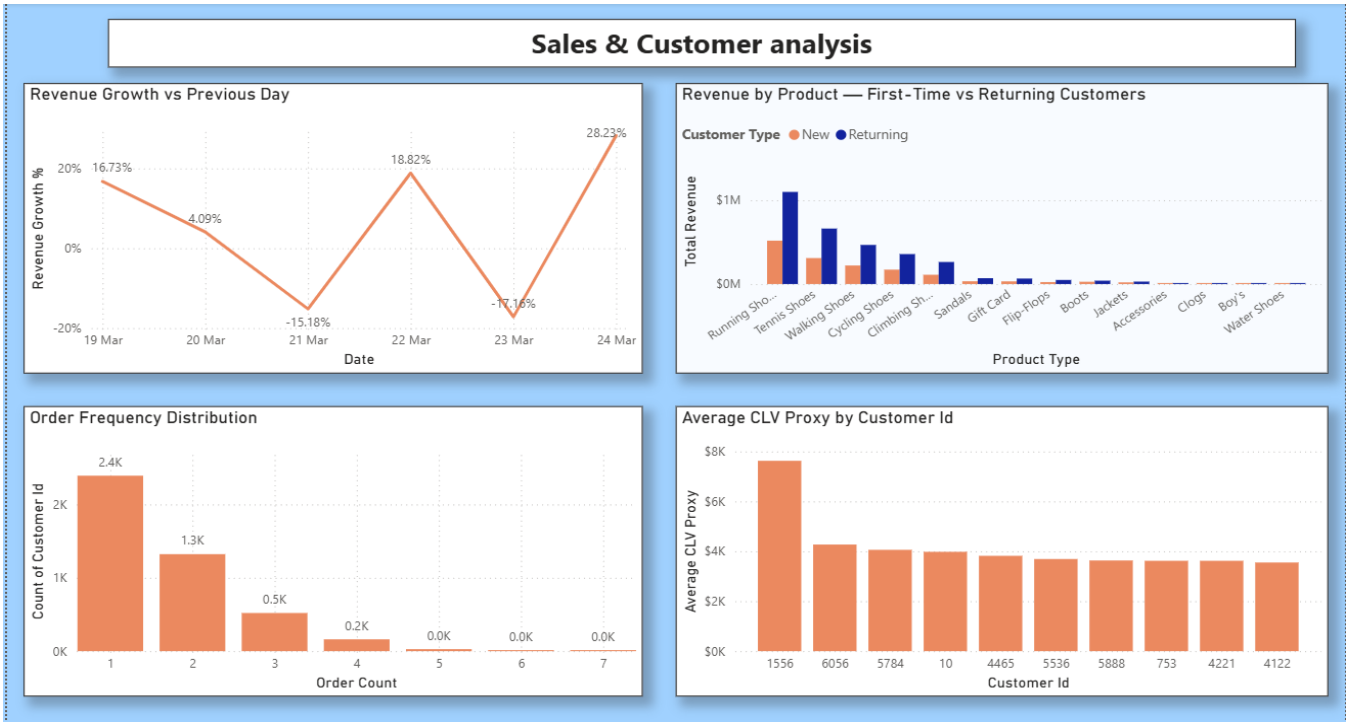


Figure 5. *Power BI Sales & Customer Analysis dashboard.*

6. MACHINE LEARNING METHODOLOGY

6.1 Predictive Problem Definition

The task is binary classification: based only on information available from a customer's first observed order, can a model estimate whether that customer places another distinct order within the following three calendar days? The target Repeat_Within_3_Days is 1 when a second distinct order occurs after the first order and within three calendar days, and 0 otherwise, provided the complete outcome window is observable. Customers whose first order occurred after 21 March 2025 were excluded because the dataset ends on 24 March and does not provide a complete three-day follow-up period for them. This produced 3,058 eligible customers, of whom 1,162 (38.00%) had a repeat order within the defined horizon.

6.2 Features, Preprocessing and Train–Test Split

The features are first-order revenue, quantity, number of product types, product type, city, province, payment gateway, weekday and order hour; later order counts, later revenue and information used to construct the target were excluded to reduce target leakage. Numeric features were median-imputed and standardised; categorical features were imputed with the most frequent value and one-hot encoded, with unseen categories handled safely. The temporal split trained the models on first-order dates 18–19 March 2025 (1,757 customers) and tested them on 20–21 March 2025 (1,301 customers).

6.3 Models

- Dummy Classifier: a majority-class baseline establishing the performance of a simple non-learning strategy.
- Logistic Regression: a linear, interpretable baseline for the repeat-purchase task.
- Random Forest Classifier: an ensemble of decision trees used to model non-linear relationships and interactions among first-order attributes.

Both learned models were trained within preprocessing pipelines so that imputation, scaling and encoding were fitted on the training data. The Random Forest used 300 trees, a minimum leaf size of five and class-weight balancing. All models were compared on the same later-date test set.

7. MODEL EVALUATION AND RESULTS

7.1 Evaluation Metrics and Model Comparison

Accuracy is the proportion of correct predictions; precision is the proportion of predicted repeat customers who actually repeated; recall is the proportion of actual repeat customers identified; F1-score is their harmonic mean; and ROC-AUC measures the ability to rank positive cases above negative cases across thresholds.

Table 6. *Model comparison on the later-date test set.*

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.48	0.36	0.48	0.41	0.49
Random Forest	0.52	0.37	0.38	0.37	0.49
Dummy (most frequent)	0.62	0.00	0.00	0.00	0.50

The Dummy Classifier achieves the highest accuracy because it predicts the majority class, but it identifies no repeat customers (zero precision, recall and F1-score for the positive class). Logistic Regression records an F1-score of 0.41 and Random Forest 0.37; both learned models have ROC-AUC values close to 0.50, indicating little ability to distinguish customers who repeat within three days from those who do not (Figure 7).

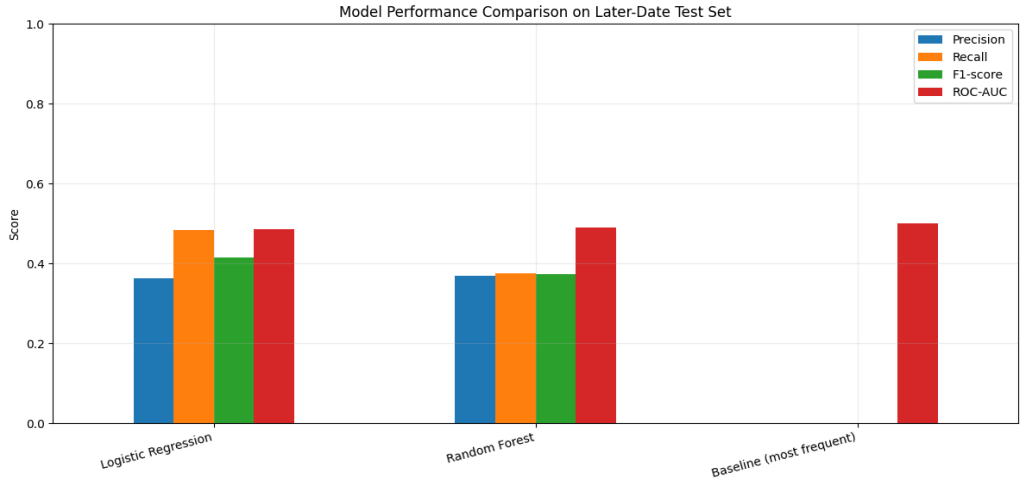


Figure 7. *Comparison of precision, recall, F1-score and ROC-AUC on the later-date test set.*

7.2 Random Forest Evaluation

The Random Forest report shows precision, recall and F1-score of 0.61 for the no-repeat class (806 test examples) and 0.37, 0.38 and 0.37 respectively for the repeat class (495 examples). Overall accuracy is 0.52 and macro-average F1-score is 0.49. The confusion matrix (Table 7, Figure 8) shows that 186 repeat customers were correctly identified, 309 were missed, and there were 317 false positive repeat predictions, so the model has limited usefulness for identifying repeat customers reliably.

Table 7. *Random Forest confusion matrix (counts).*

Actual / predicted	Predicted: No repeat	Predicted: Repeat
Actual: No repeat	489	317
Actual: Repeat	309	186

7.3 ROC, Precision–Recall and Feature Importance

The ROC curve has an area under the curve of approximately 0.49, close to chance-level ranking, and the precision–recall curve has average precision of approximately 0.37 (Figures 9 and 10). These plots support the conclusion that

the model does not learn a strong predictive signal from the available first-order attributes and short observation window.

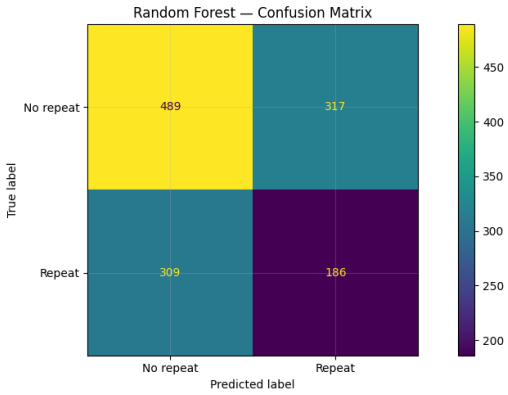


Figure 8. *Random Forest confusion matrix.*

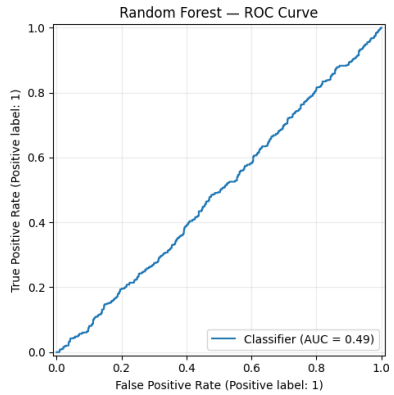


Figure 9. *ROC curve (AUC ≈ 0.49).*

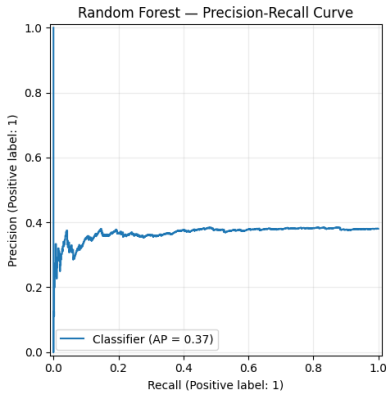


Figure 10. *Precision–recall curve (AP ≈ 0.37).*

Permutation feature importance (Figure 11) measured the change in test F1-score when individual features were shuffled. Province and product type show positive importance in this run, while several other features have small or negative estimated contributions. Because overall model performance is weak and permutation estimates can vary, these values are exploratory and should not be treated as reliable business drivers.

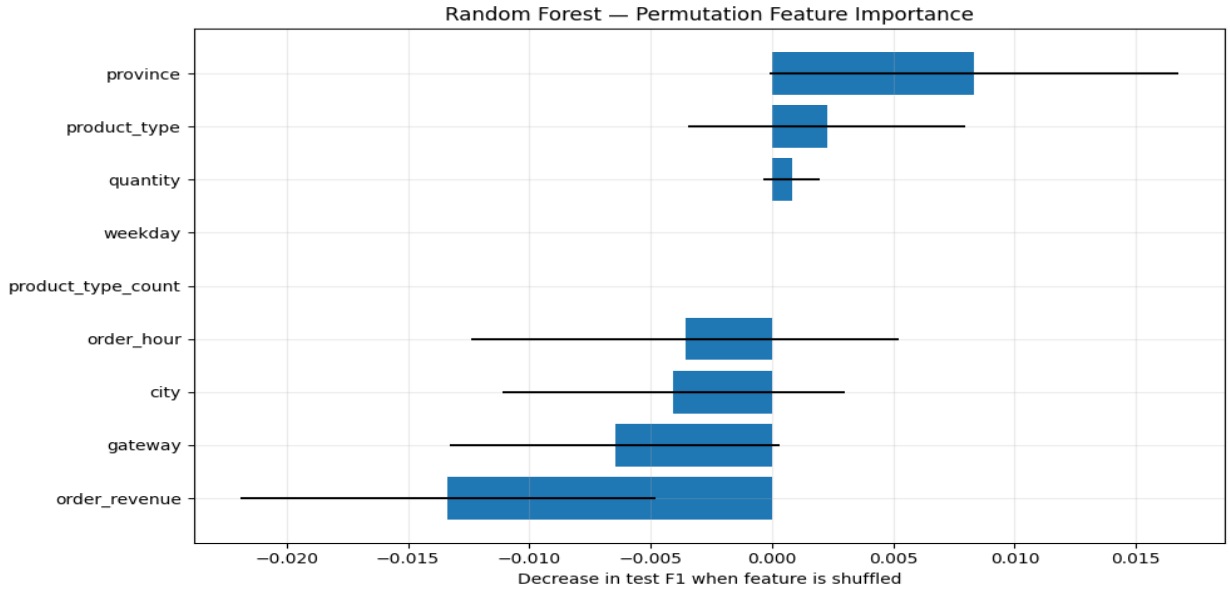


Figure 11. *Random Forest permutation feature importance.*

7.4 Interpretation of Results

The experiment does not provide evidence of reliable three-day repeat-purchase prediction. The short seven-day period restricts the number of customers with fully observed follow-up windows and limits the historical context available to the models. The results should therefore be reported as a negative or inconclusive predictive finding. The descriptive Power BI analysis remains useful for summarising the supplied transactions, but it should not be confused with a validated long-term retention model.

8. KEY FINDINGS AND BUSINESS INSIGHTS

- The dataset contains 7,431 orders and 4,431 unique customers over 18–24 March 2025.

- Total revenue is USD 4,598,960.87 and average order value is USD 618.89.
- Running Shoes is the largest product category by revenue at USD 1,616,946.28, or 35.16% of total revenue.
- The observed-period repeat-customer rate is 46.02%; one-time customers represent 53.98% of unique customers.
- The Power BI analysis reports an average first-to-second purchase gap of 2.03 days among customers with a second observed purchase.
- The top 10% of customers contribute 22.05% of revenue, highlighting the importance of understanding higher-value customers.
- The Random Forest model achieves F1-score 0.37 and ROC-AUC 0.49 on the later-date test set; this is weak predictive performance.

9. BUSINESS RECOMMENDATIONS

- **Customer retention:** Treat the observed 2.03-day purchase gap as a hypothesis for testing timely post-purchase engagement (relevant recommendations, order-support messages or follow-up offers, subject to customer consent and communication policies). It should not be treated as a universal optimum and should be validated on a longer dataset or through a controlled campaign experiment.
- **Product and inventory:** Review inventory availability and replenishment for the leading categories (Running Shoes, Tennis Shoes and Walking Shoes), considering units sold, margins, stock levels and supplier lead times, which revenue ranking alone does not establish.
- **Marketing and engagement:** Use customer-level revenue and purchase frequency to form practical customer groups without assuming lifetime value from a one-week sample, and evaluate loyalty incentives and cross-selling through measurable tests that track second-purchase conversion, repeat-purchase rate and revenue per customer over a longer period.

10. LIMITATIONS AND FUTURE ENHANCEMENTS

10.1 Limitations

- **Dataset:** The dataset covers only seven days (18–24 March 2025) and cannot support robust seasonal analysis, month-over-month comparisons or long-term cohort retention. Customers who purchase after the observation period may appear as one-time customers, so the 2.03-day purchase gap and 46.02% repeat rate are observed-period measures, not lifetime behaviour estimates.
- **Machine learning:** Only 3,058 customers had a complete three-day outcome window, and the temporal test set contains 1,301 customers. Random Forest ROC-AUC is approximately 0.49, indicating that the selected first-order features do not provide useful ranking performance in this experiment; the models should not be deployed for automated customer targeting based on these results.

10.2 Future Scope

- Extend the dataset to several months or years and include acquisition channel, discounts, returns, margins, inventory and campaign interactions where available.
- Evaluate longer prediction horizons and survival/time-to-event approaches for repeat-purchase timing.
- Use cross-validation or rolling temporal validation across multiple periods, and test calibrated models and business thresholds only after predictive performance improves on unseen data.
- Compare customer-engagement strategies through controlled experiments and monitor incremental repeat purchases.

11. CONCLUSION

This project demonstrates an end-to-end analytical workflow for Shopify sales and customer data, combining Power BI dashboards with Python-based exploratory analysis and machine learning. The descriptive analysis identifies USD 4.60 million in revenue, 7,431 orders, 4,431 customers and an average order value of USD 618.89, and highlights the strong revenue contribution of Running Shoes and the observed balance between one-time and repeat customers.

The machine-learning experiment compared Logistic Regression, Random Forest and a majority-class baseline for predicting a second order within three days of a customer's first observed purchase. Random Forest achieved an F1-score of 0.37 and ROC-AUC of 0.49, showing that the available data does not support reliable prediction. Longer-term transaction history and richer customer attributes are necessary before predictive outputs can guide retention decisions. Overall, descriptive analytics supports business understanding, while careful model evaluation prevents unsupported predictive claims.

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